LAB SHEET 1

Full Stack Development — Mini E-Commerce Project

Building the Storefront (HTML + CSS)
Covers Project Phase 1 & Phase 2

Objective	Build the static page structure of the e-commerce store, then style it into a responsive storefront.
Duration	2 hours
Prerequisite	Group formed, product category registered
Submission	index.html, product.html, cart.html, checkout.html + style.css

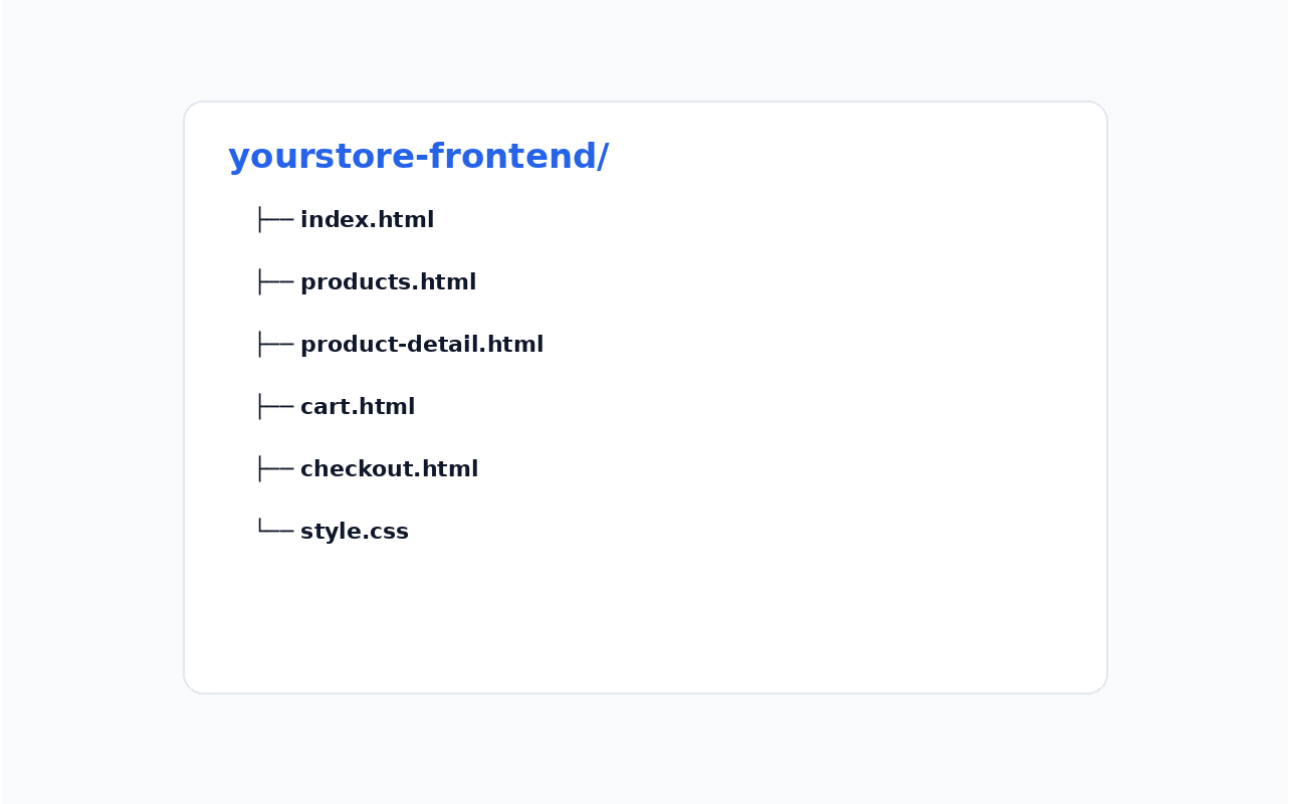
Completed solution visuals and answers

The following pages answer all 16 tasks from the supplied Lab Sheet 1 and show a corresponding output or diagram for each task.

Question 1: Create the project folder and required pages.

Answer: Create a folder named **yourstore-frontend** and add **index.html**, **products.html**, **product-detail.html**, **cart.html**, **checkout.html**, plus **style.css**.

```
yourstore-frontend/
■■■ index.html
■■■ products.html
■■■ product-detail.html
■■■ cart.html
■■■ checkout.html
■■■ style.css
```

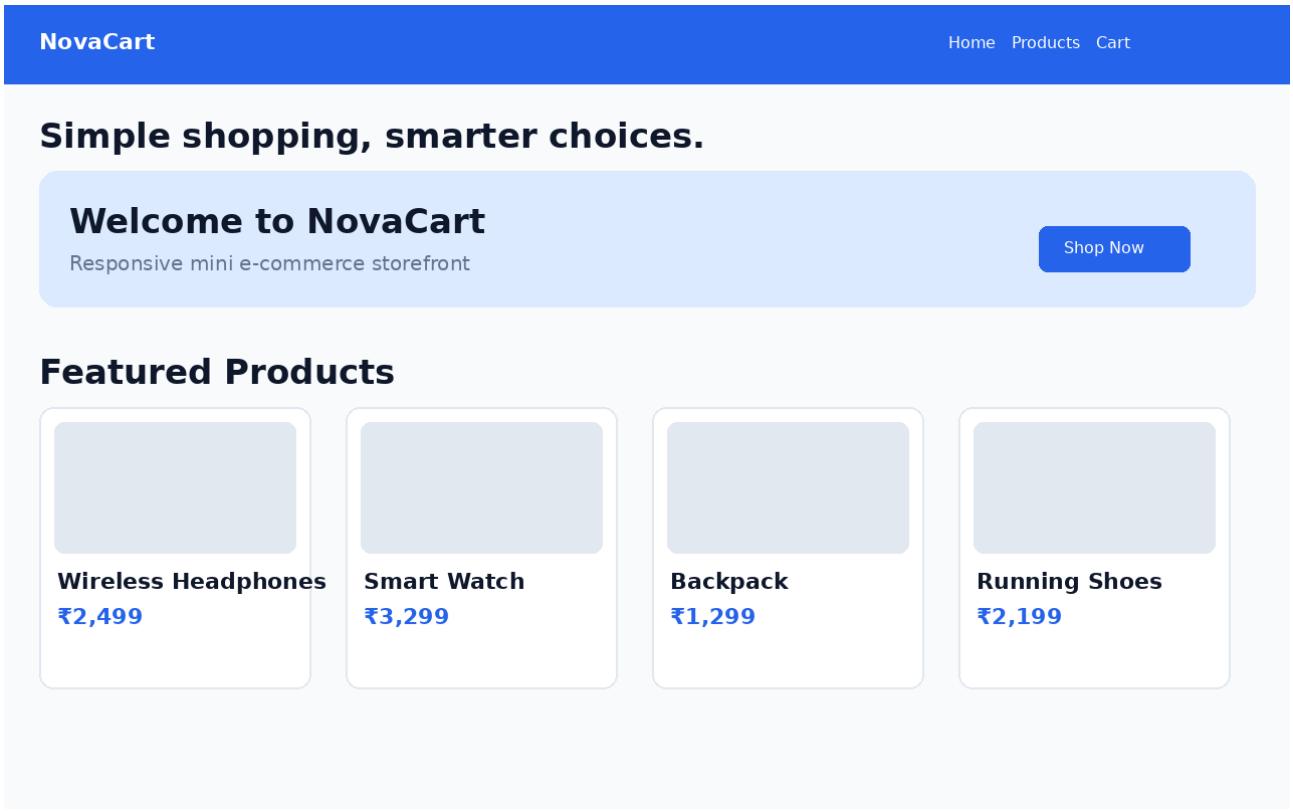


Output / diagram for Question 1: Project folder and file structure.

Question 2: Add a common header and navigation on every page.

Answer: Use a shared **<header>** containing the store name/logo and a **<nav>** with links to Home, Products and Cart. The same navigation is used throughout the storefront.

```
<header>
<div class="logo">NovaCart</div>
<nav><a href="index.html">Home</a> <a href="products.html">Products</a> <a href="cart.html">Cart</a></nav>
</header>
```

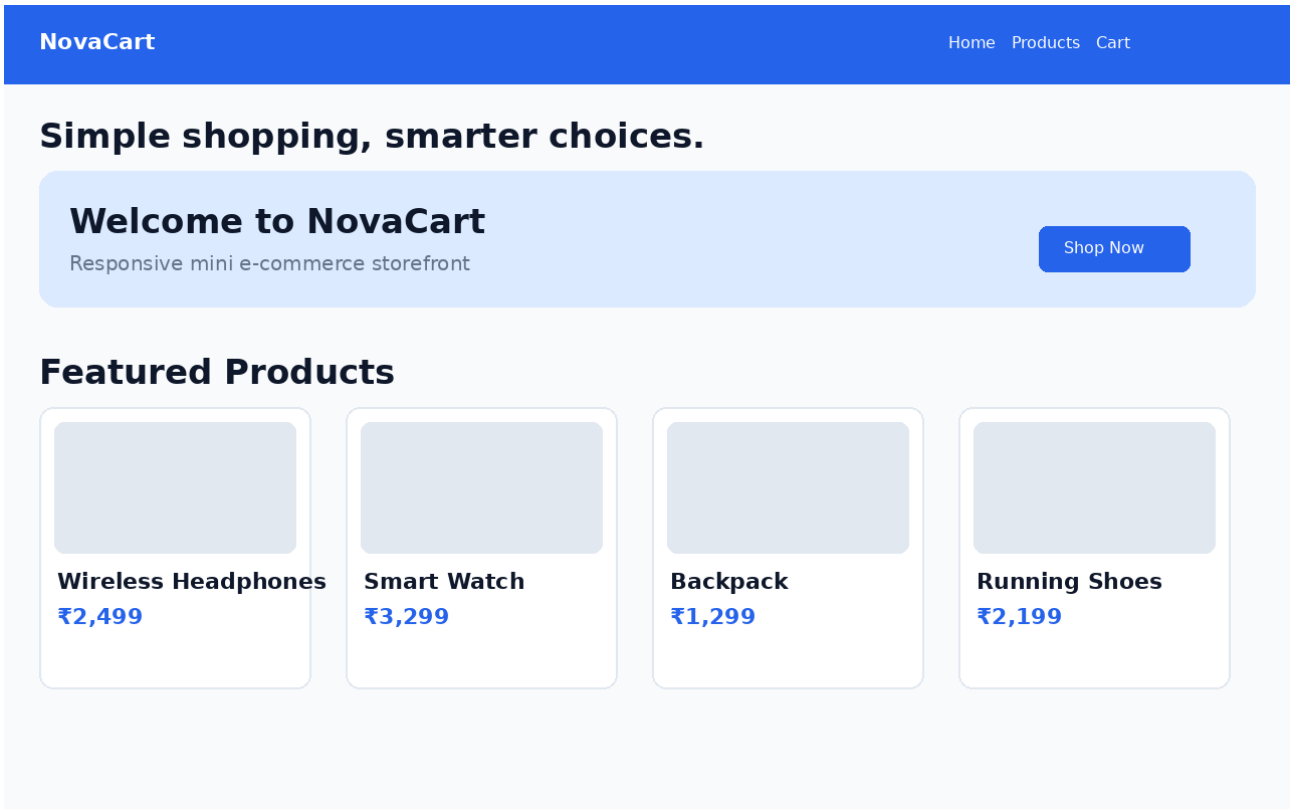


Output / diagram for Question 2: Common header and navigation output.

Question 3: Build index.html with a hero/banner and at least 4 Featured Products.

Answer: The home page contains a hero/banner section introducing the store and four placeholder product blocks, each showing an image area, product name and price.

```
<section class="hero">...store introduction...</section>
<section class="grid">
<!-- 4 featured product cards -->
</section>
```

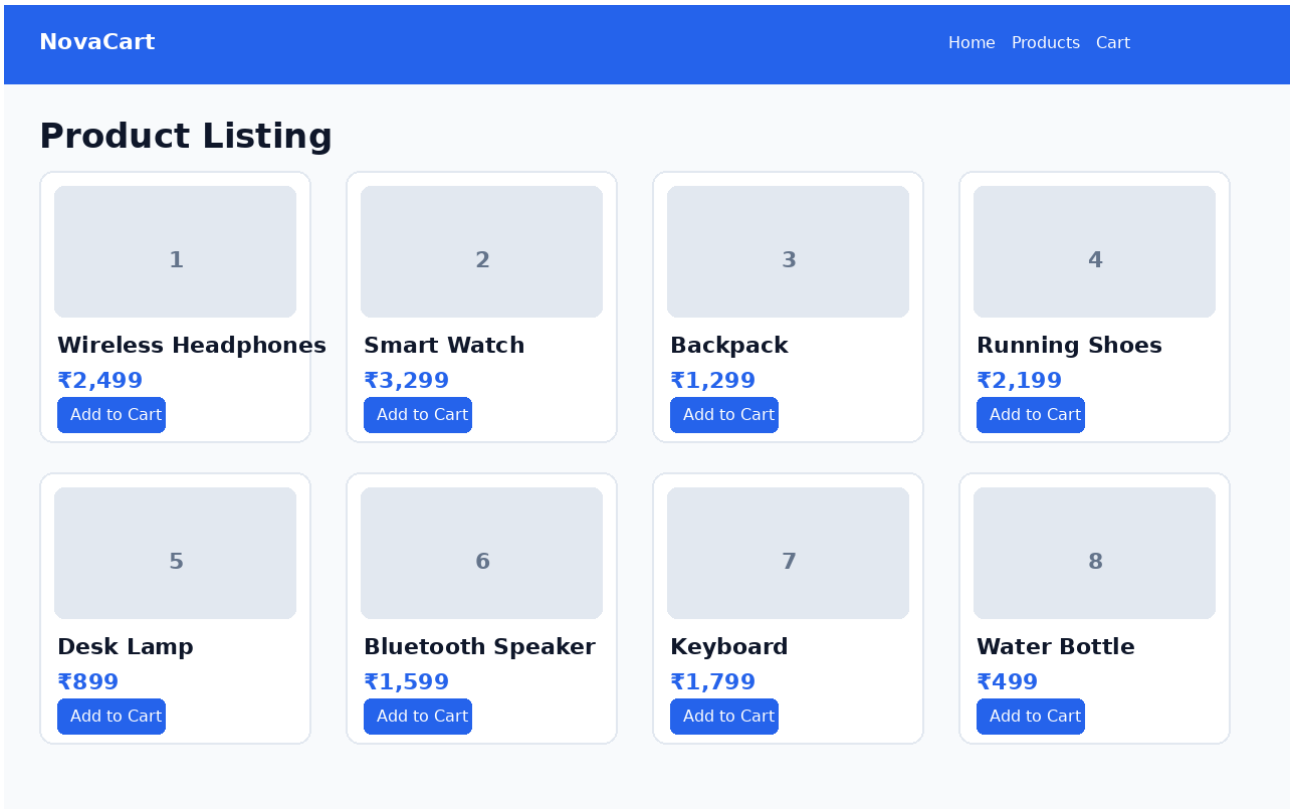


Output / diagram for Question 3: Home page output with four featured products.

Question 4: List at least 8 products on products.html.

Answer: Create eight repeated product cards. Each card contains an image area, product name, price and an **Add to Cart** button.

```
<div class="card">
<div class="thumb">Image</div>
<b>Product Name</b><p class="price">■2,499</p>
<button class="btn">Add to Cart</button>
</div>
```

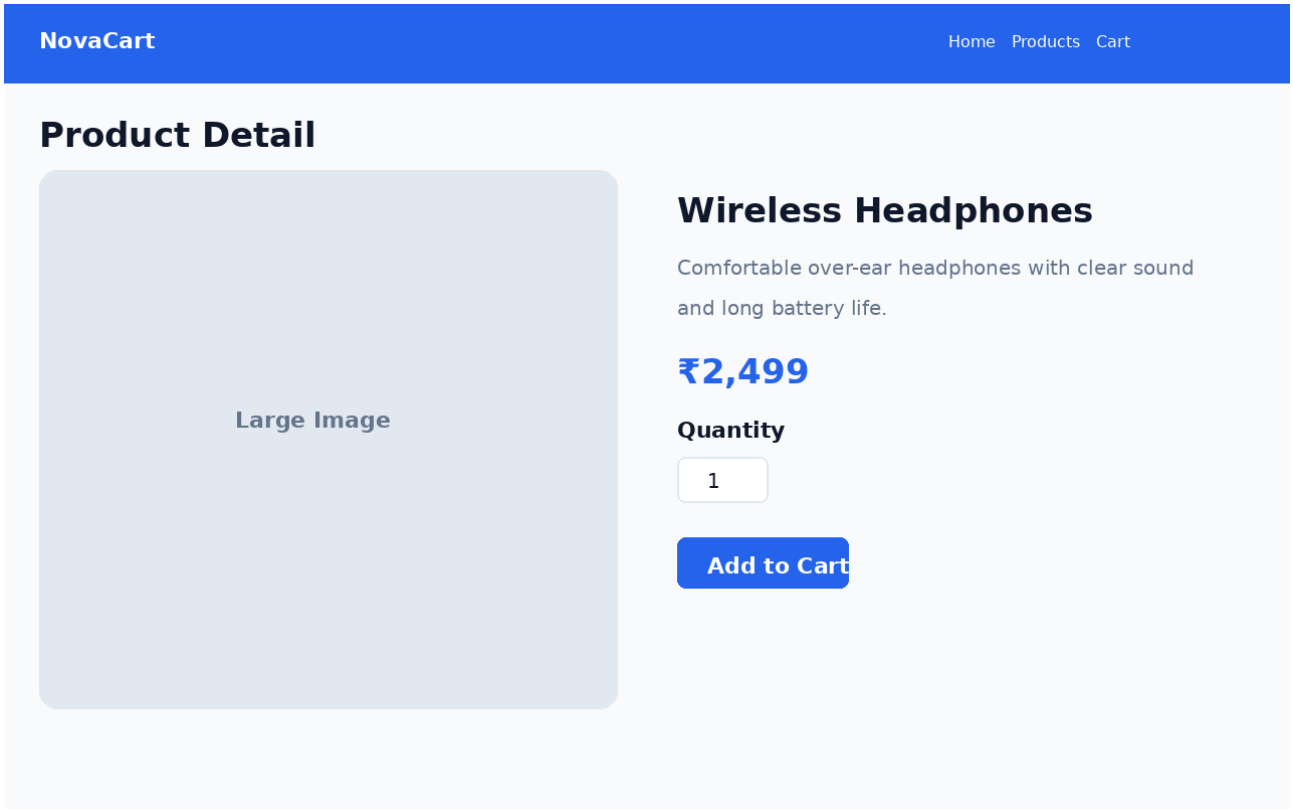


Output / diagram for Question 4: Product listing output with 8 cards.

Question 5: Show full product details on product-detail.html.

Answer: Display a larger product image, name, description, price, quantity selector and **Add to Cart** button.

```
<div class="detail">
<div class="thumb">Large Product Image</div>
<div><h1>Wireless Headphones</h1>...<select class="qty">...</select>...</div>
</div>
```

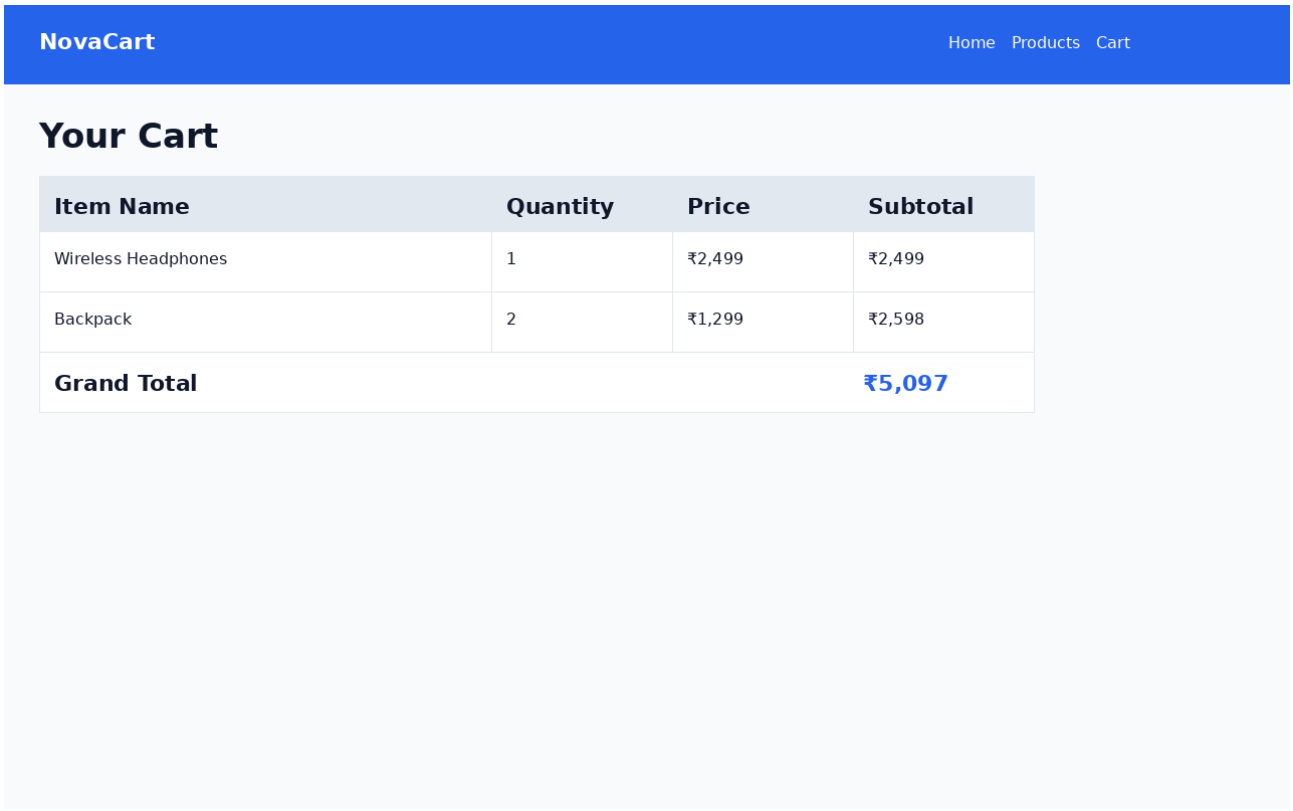


Output / diagram for Question 5: Product detail page output.

Question 6: Create the cart table/list with subtotal and grand total.

Answer: The cart shows item name, quantity, price and subtotal, followed by a grand-total row. Example total shown: **■5,097**.

```
<table>
<tr><th>Item Name</th><th>Quantity</th><th>Price</th><th>Subtotal</th></tr>
...
<tfoot><tr><td colspan="3">Grand Total</td><td>■5,097</td></tr></tfoot>
</table>
```

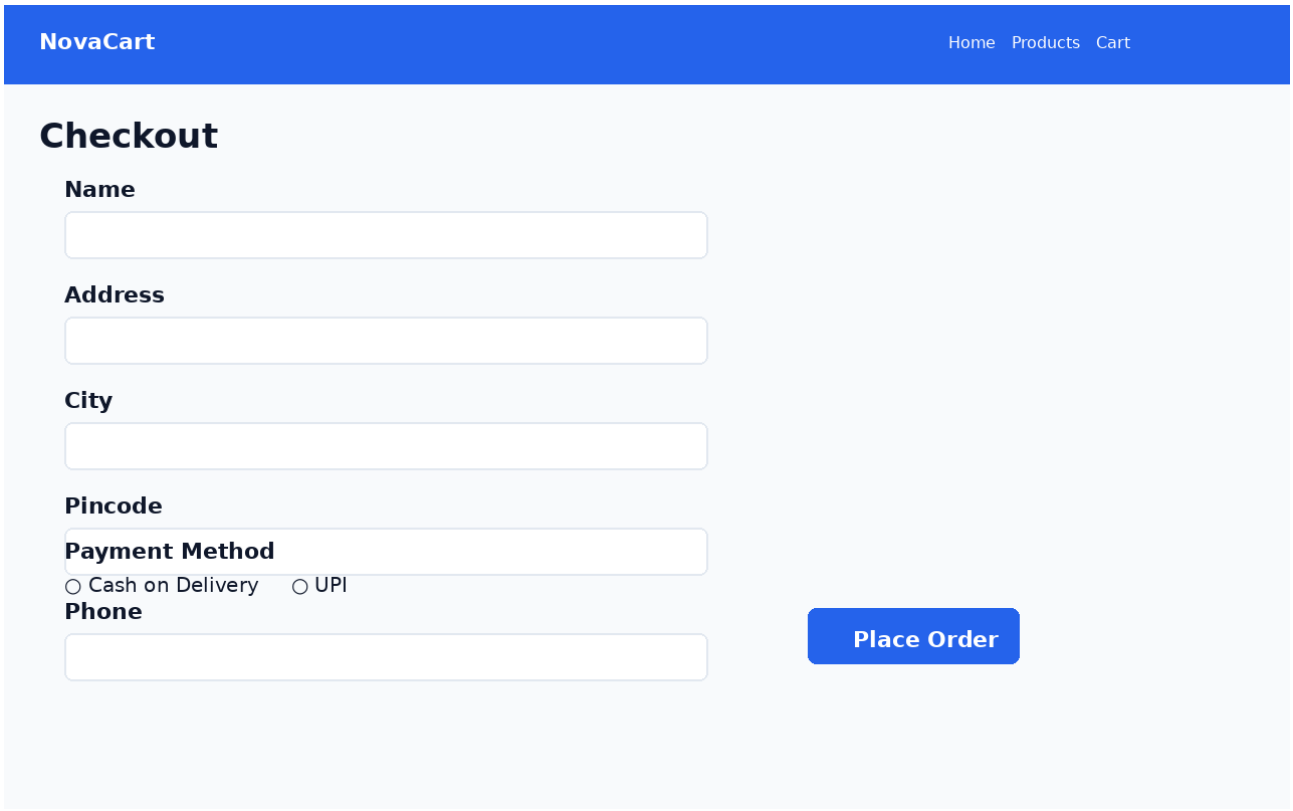


Output / diagram for Question 6: Cart table output.

Question 7: Create the checkout form.

Answer: The form contains Name, Address, City, Pincode, Phone and Payment Method radio buttons, followed by a **Place Order** button.

```
<form class="form">
Name, Address, City, Pincode, Phone
Payment Method: radio buttons
<button class="btn">Place Order</button>
</form>
```

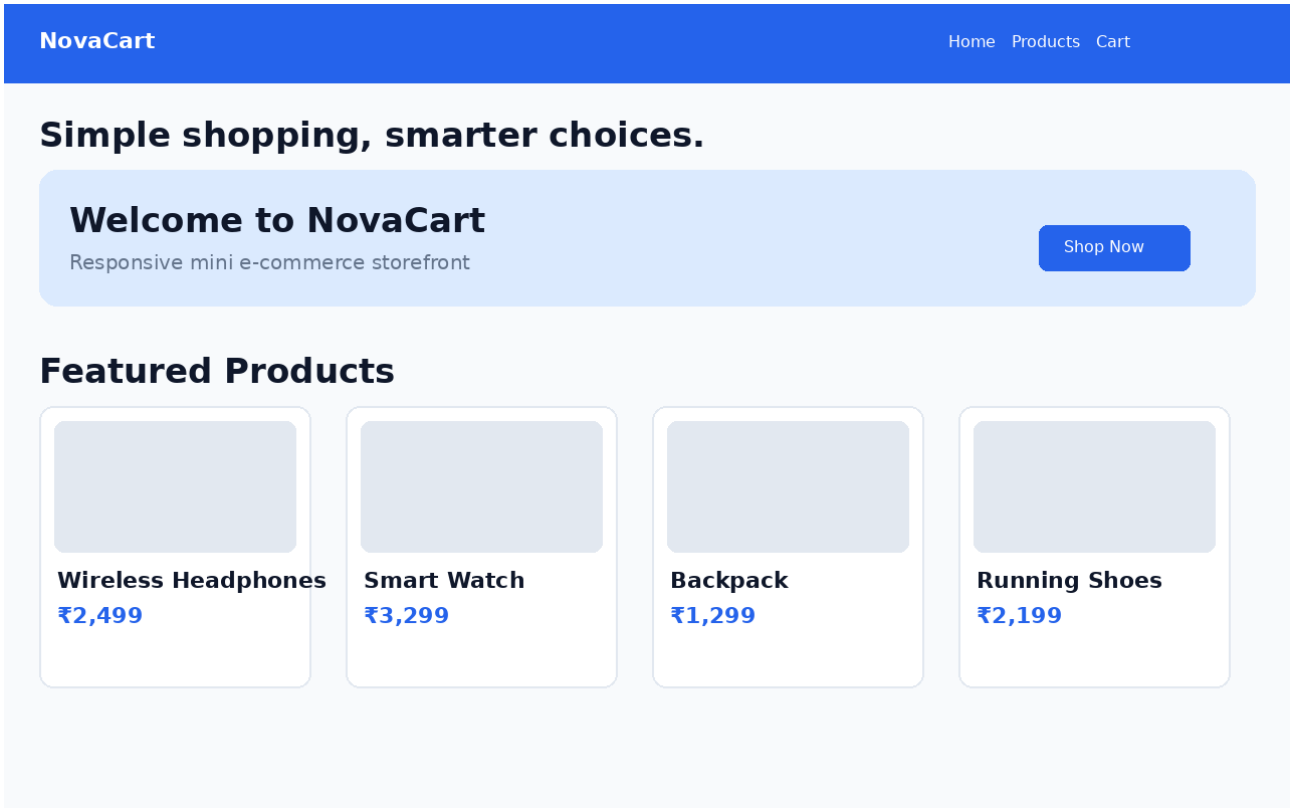


Output / diagram for Question 7: Checkout form output.

Question 8: Add a common footer to every page.

Answer: Use a shared **<footer>** containing contact information and copyright text. The footer is visually consistent across pages.

```
<footer>Contact: support@novacart.test · © 2026 NovaCart</footer>
```

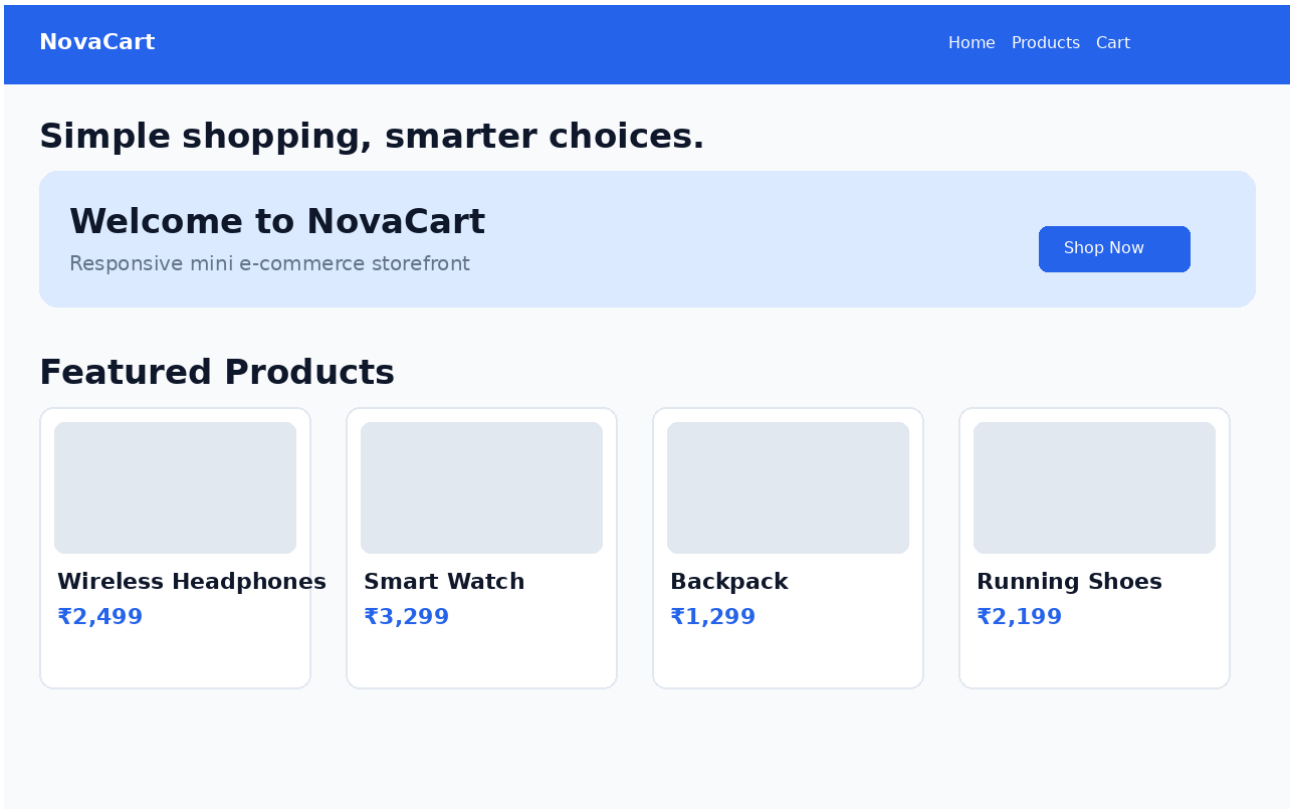


Output / diagram for Question 8: Footer visible at the bottom of the storefront.

Question 9: Create style.css and link it to all 5 pages.

Answer: Add **<link rel="stylesheet" href="style.css">** inside the **<head>** of each page. This gives all five pages the same styling system.

```
<head>
...
<link rel="stylesheet" href="style.css">
</head>
```

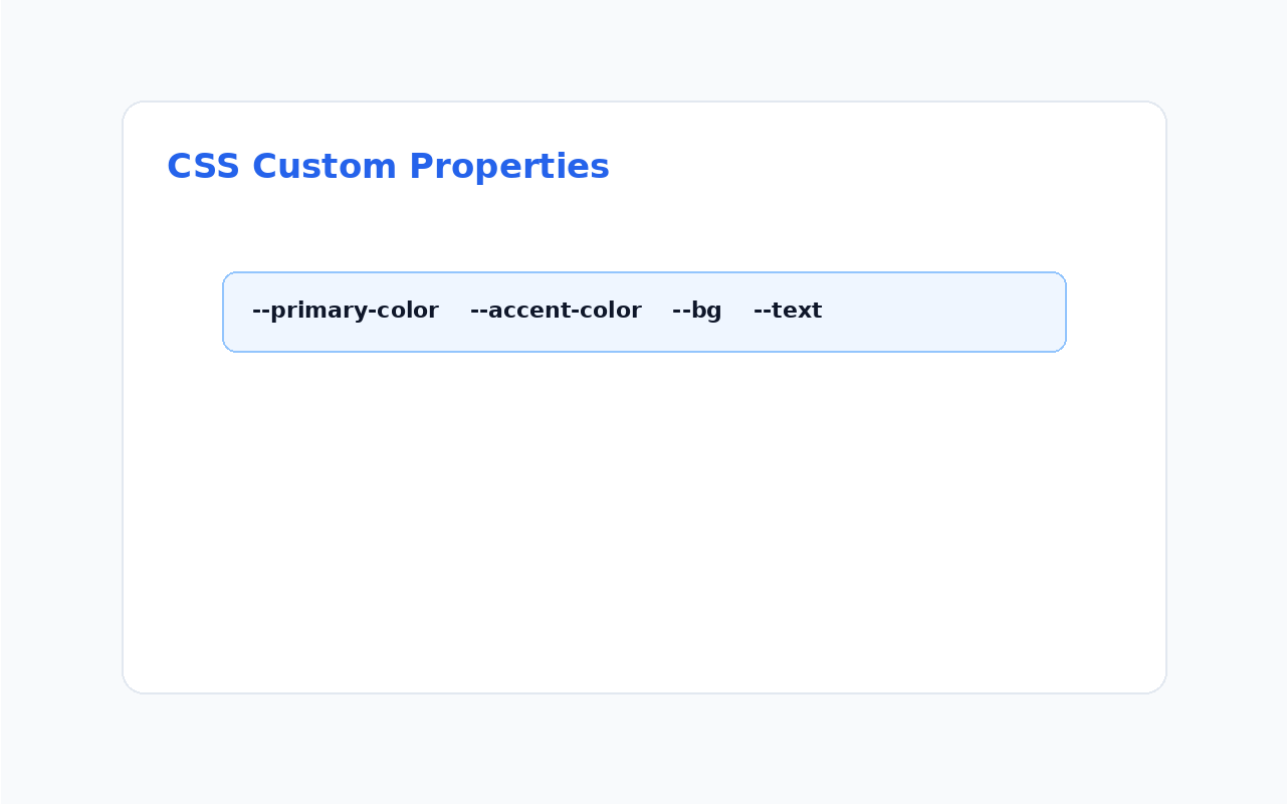


Output / diagram for Question 9: Styled page output demonstrating the linked stylesheet.

Question 10: Define 3–4 CSS custom properties for the color theme.

Answer: Define theme variables in **:root**, for example **--primary-color**, **--accent-color**, **--bg** and **--text**, then reuse them in the stylesheet.

```
:root {
--primary-color: #2563eb;
--accent-color: #f59e0b;
--bg: #f8fafc;
--text: #0f172a;
}
```

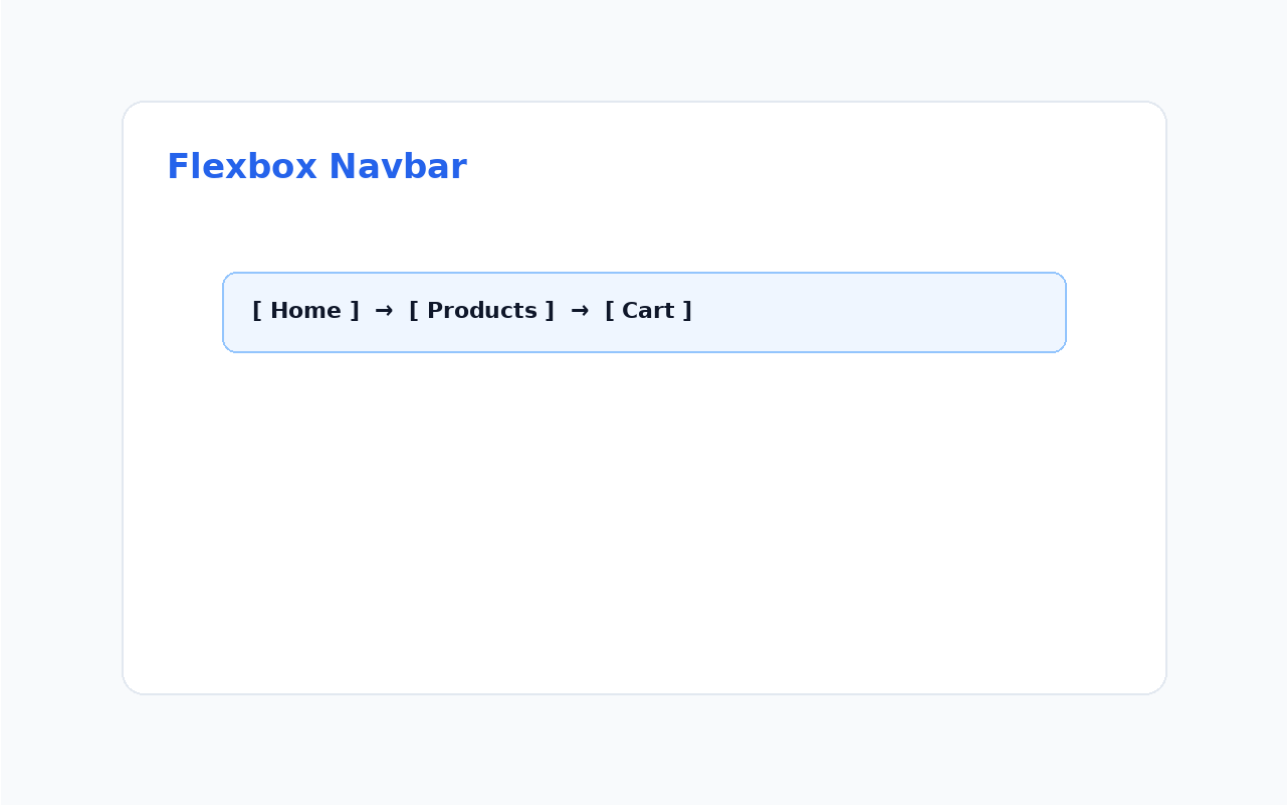


Output / diagram for Question 10: CSS custom-properties theme diagram.

Question 11: Style the navbar using Flexbox.

Answer: Apply **display: flex** to the navigation, with spacing between links. A media query changes the layout for smaller widths so the links can stack vertically.

```
nav {
display: flex;
gap: 24px;
}
@media(max-width:700px){ nav{flex-direction:column;} }
```

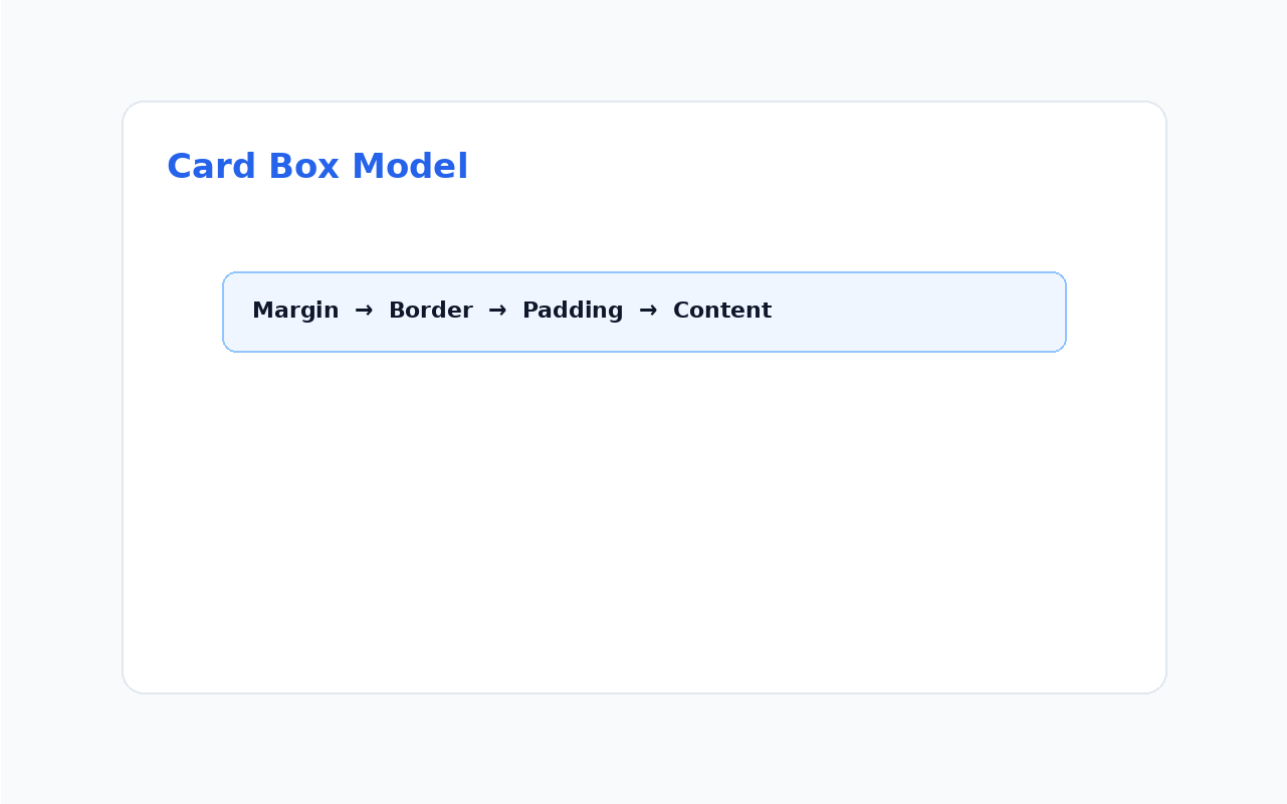


Output / diagram for Question 11: Flexbox navbar diagram.

Question 12: Style product blocks as cards using Box Model concepts.

Answer: Product cards use padding, border, border-radius and box-shadow. These properties demonstrate the CSS Box Model while creating a consistent card appearance.

```
.card {
padding: 16px;
border: 1px solid #e2e8f0;
border-radius: 14px;
box-shadow: 0 5px 14px rgba(15,23,42,.07);
}
```



Output / diagram for Question 12: Box Model diagram for a product card.

Question 13: Arrange the product listing responsively using Flexbox or CSS Grid.

Answer: Use CSS Grid for the product listing so cards repeat across the row. The layout wraps naturally and changes to a single column at small widths.

```
.grid {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 18px;
}
@media(max-width:700px){ .grid{grid-template-columns:1fr;} }
```

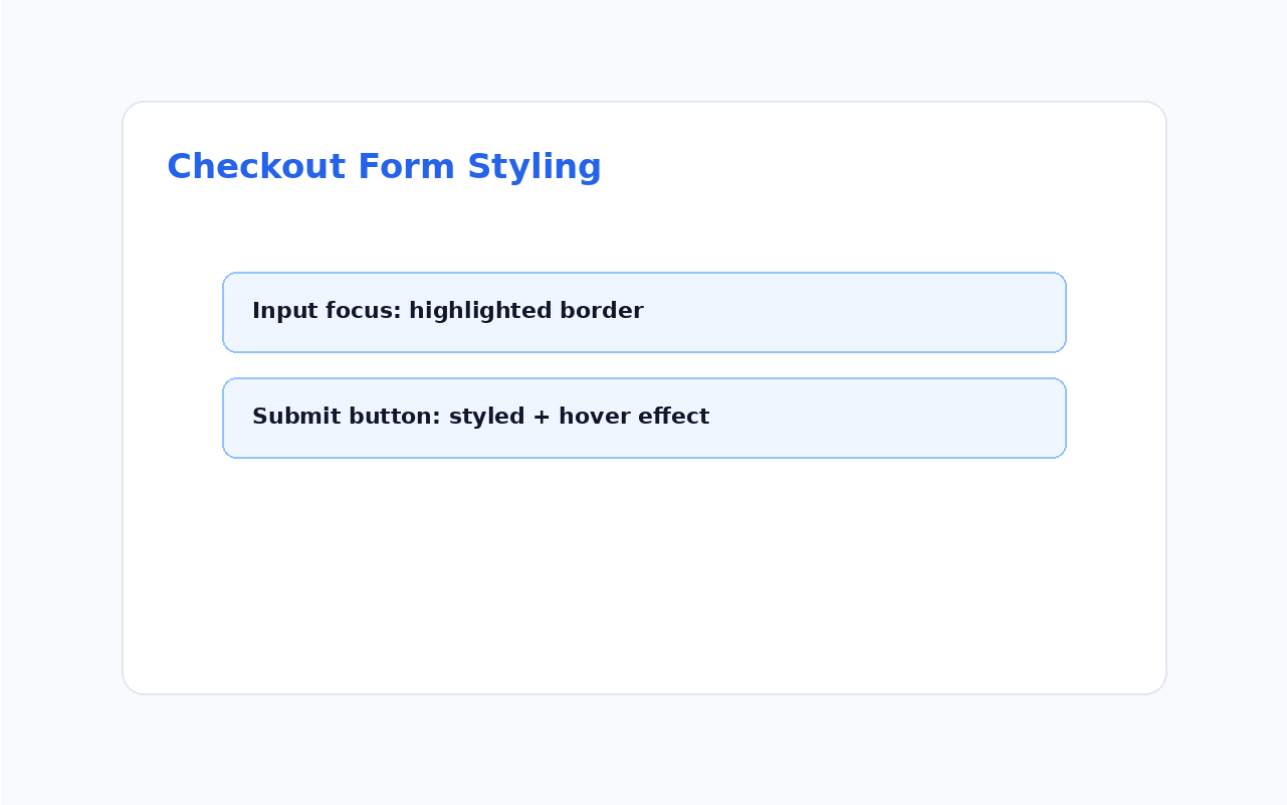


Output / diagram for Question 13: Responsive product-grid diagram.

Question 14: Style the checkout form with focus and hover states.

Answer: Inputs have consistent sizing, padding and a visible focus state. The **Place Order** button is styled and prepared with a hover effect.

```
.form input:focus {
outline: 2px solid #93c5fd;
border-color: var(--primary-color);
}
.btn:hover { filter: brightness(.9); }
```

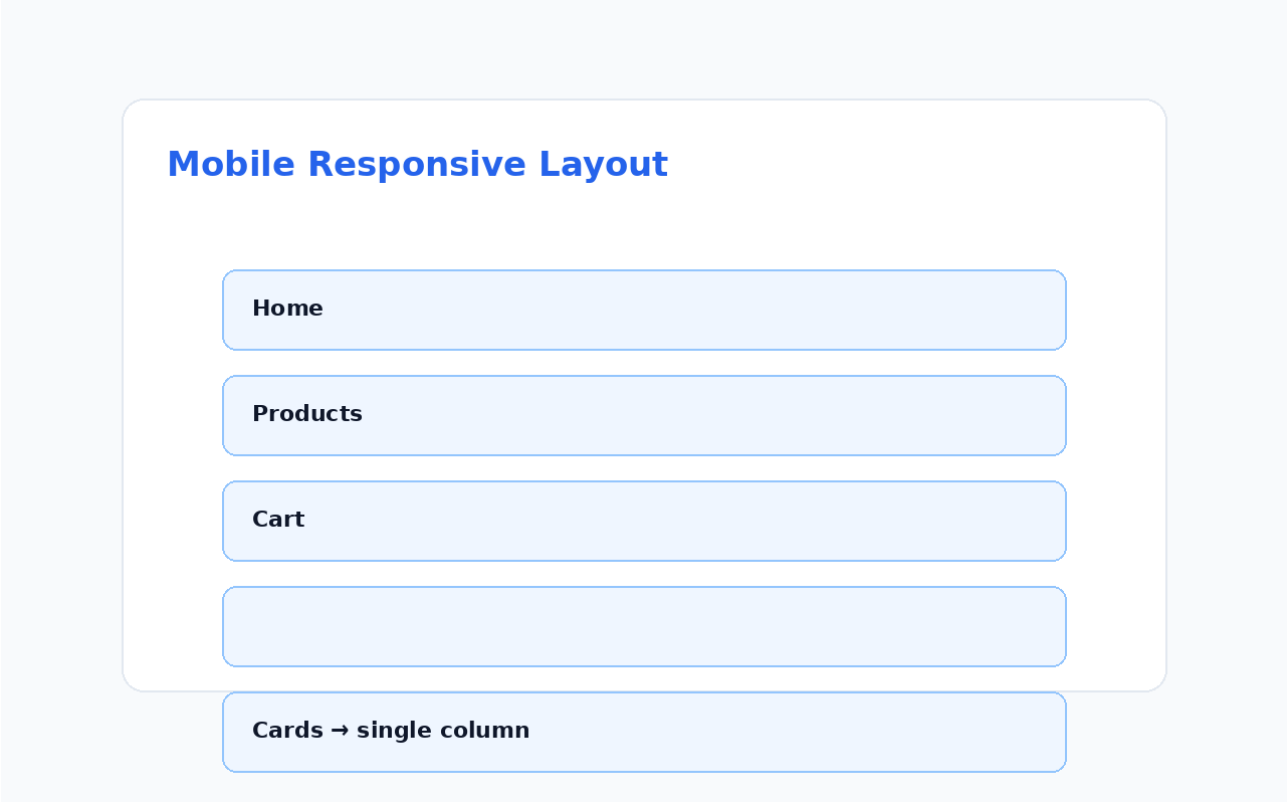


Output / diagram for Question 14: Checkout form styling diagram.

Question 15: Add at least one @media query for mobile width.

Answer: The media query changes navigation to a vertical stack, makes product cards single-column, and adjusts the detail layout for smaller screens.

```
@media(max-width:700px) {
nav { flex-direction: column; }
.grid { grid-template-columns: 1fr; }
.detail { grid-template-columns: 1fr; }
}
```

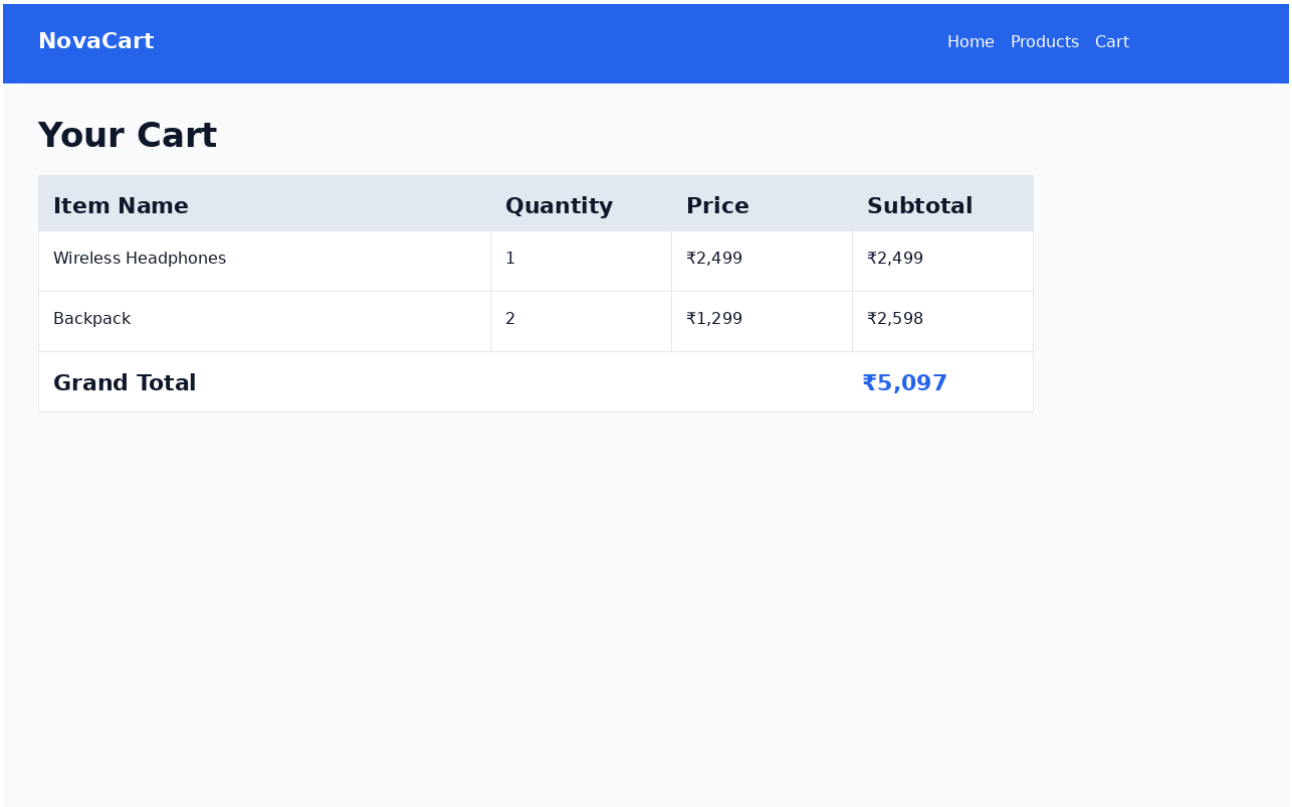


Output / diagram for Question 15: Mobile-responsive layout diagram.

Question 16: Style the cart and footer to match the overall theme.

Answer: The cart table uses the same spacing, borders and theme colors as the cards and form. The footer uses the same overall visual language and is repeated on every page.

```
table { width:100%; border-collapse:collapse; }
th, td { padding:14px; border-bottom:1px solid #e2e8f0; }
footer { background:#0f172a; color:#fff; padding:24px; }
```



Output / diagram for Question 16: Cart styling output matching the storefront theme.

Submission Checklist

✓ All five HTML pages created and linked through the navbar.

✓ Home page contains hero/banner and 4 featured products.

✓ Product listing contains at least 8 products.

✓ Product detail includes image, description, price, quantity selector and Add to Cart.

✓ Cart contains quantity, price, subtotal and grand total.

✓ Checkout contains all requested fields and payment radio buttons.

✓ Common footer appears on every page.

✓ style.css is linked to all pages and uses CSS variables.

✓ Navbar uses Flexbox and the product listing uses a responsive grid.

✓ Cards, checkout form, cart and footer are styled consistently.

✓ At least one @media query adapts the layout for mobile widths.

Source basis

Prepared directly from the supplied “BTCS303P-AS01” Lab Sheet 1: Building the Storefront (HTML + CSS). The source specifies the objective, submission items, Part A HTML tasks 1–8, and Part B CSS tasks 9–16.